Flight Simulator Platform and Force Feedback Peripherals

A Graduation Project
Submitted in Partial Fulfilment of B.Sc. Degree Requirements in Mechatronics System Engineering

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Introduction

Flight simulators nowadays are not only used by professionals for training in the civil and military fields, but the availability of flight simulation software such as Microsoft's Flight Simulator and X-plane, encouraged many individuals and companies to create renditions of flight simulators in order to enhance the user experience. The proposed project aims to design and implement a flight simulator platform complete with pilot controls for the purposes of early training and recreational video-game usage by hobbyists. This will alleviate the high cost on flight training academies and individuals who are in the market for a simulation platform. The project will also tackle motion cueing, which is used to relay the correct motion feedback to the user onboard.

Objectives

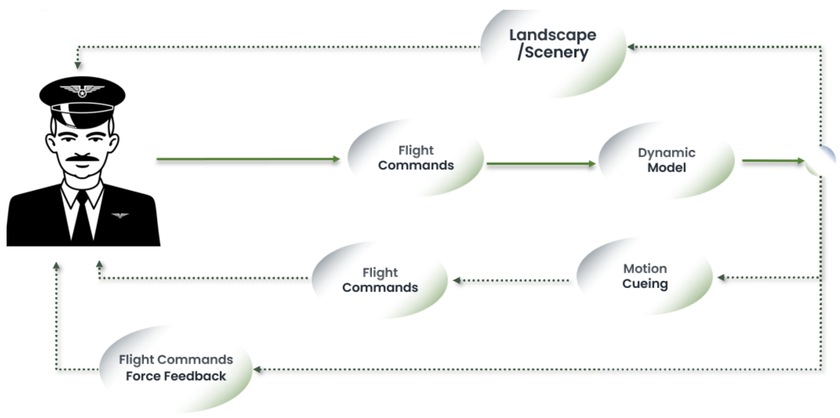
To undertake the design and hardware implementation of a human-in-the-loop flight simulator complete with peripherals and a motion platform to **Circumvent** the excessive costs associated with purchasing a platform with similar features.

Results

The systems developed were analyzed, including interfacing, motion cueing algorithms, and motion platform performance. Results showed promising progress in the development of motion cueing algorithms, with the optimal washout filter outperforming the classical approach. The motion platform also showed significant progress in acceleration compared to the aircraft, evaluated based on user experience. These results are promising for future advancements in the field.

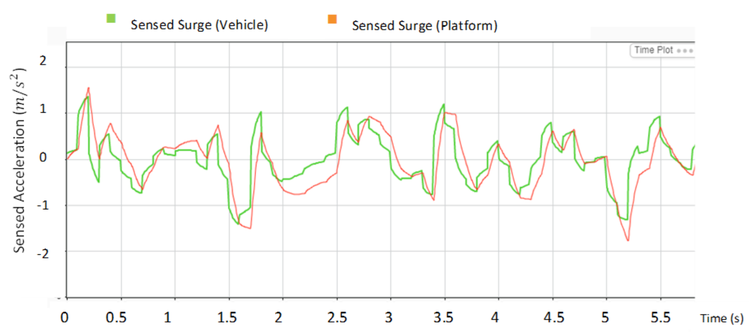
Pilot-In-the Loop:

First microcontrollers that are responsible for receiving the flight commands from the pilot are connected to MATLAB through serial communication. a SIMULINK model is responsible for receiving the data from the microcontroller and then send the variables to X-plane software through User Datagram Protocol (UDP) communication. After sending the flight commands to the simulation environment (X-plane), it sends the visual and audio cues to the displays and the speakers. X-plane calculates the dynamic model of the aircraft then sends it back to MATLAB. After receiving the dynamic model data, a SIMULINK model responsible for calculating the required accelerations and velocities to deliver the same sensation to the pilot (motion cueing), this data should be fed to a system that calculates the inverse kinematics of the platform, this communication is done through UDP communication. On the other hand, the data extracted from the simulation environment will help in creating the haptic feedback to the input peripherals. These data are fed to the microcontrollers to control the actuators that are responsible for generating the force feedback. After calculating the inverse kinematics for the platform, the required motor angles are fed to the motion platform microcontroller to control the actuators.



Optimal Motion Cueing Algoritm

The optimal control-based algorithm approaches motion cueing as a tracking problem, with the goal of minimizing the sensation error between the sensed accelerations on the platform and in the real vehicle. Higher order filters with optimal control strategies, namely the linear quadratic regulator, LQR, are used in this algorithm.



Motion Platform

A 2-DOF (pitch and roll) motion platform is a type of simulator that allows users to experience the sensation of flying an aircraft or driving a vehicle by providing motion feedback based on the movements of the user's input devices. The 2-DOF motion rig is designed to simulate the pitch and roll movements of the aircraft, which are the primary motions that affect the pilot's perception of flight.

Conclusions

The Flight Simulator project successfully achieved its objectives, including the design and implementation of pilot control peripherals, motion cueing algorithms, and a 2 DOF motion platform. The project also explored advanced motion cueing algorithms and added force feedback to the yoke pilot control peripheral. The related paper was accepted for the AISE Conference. The project's progress has laid the groundwork for future advancements in the field.

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